

Lab: Pragmatism and the Primacy of Practice

Learning Goal: Apply pragmatist philosophy to Active Inference's account of action.

Part 1: Pragmatist Mapping

Exercise: Complete the mapping and evaluate each:

Pragmatist Concept	Active Inference Analog	Evaluation
Peirce: Doubt triggers inquiry	Prediction error triggers inference	
James: Truth = what works	Model evidence = predictive success	
Dewey: Cognition as instrument	Generative model as survival tool	
Dewey: Habit as organized activity	Policy prior	
Peirce: Abduction	Bayesian model selection	
Mead: Self as social construction	Self-model via interpersonal inference	

Which mapping is strongest? Which is most misleading? Support your claim with a specific example where the mapping breaks down. (300 words)

Write your response here

Part 2: Affordance Analysis

Learning Goal: Apply affordance theory through Active Inference formalization.

Exercise: Analyze the affordances of a simple object (e.g., a chair) for different agents:

Agent	Affordance of a Chair	EFE Analysis (why this affordance?)
Human adult	Sitting (reduces physical fatigue PE)	
Toddler	Climbing (reduces curiosity PE)	
Cat	Platform (reduces surveillance PE)	
Robot vacuum	Obstacle (increases navigation PE)	
Person with disability	Barrier or aid (depends on design — precision on mobility model)	

How does the affordance depend on the agent's generative model? What determines whether the same object affords different actions for different agents? Connect your answer to the precision-weighted policy evaluation in EFE.

Write your response here

Part 3: Skill Acquisition Diary

Learning Goal: Apply Dreyfus's expertise model through Active Inference.

Exercise: Choose a skill you've developed (playing an instrument, coding, cooking, a sport). Map your progression through Dreyfus's five stages:

Stage	Your Experience	Generative Model Status	Policy Prior Status	Precision Dynamics
Novice		Thin, rule-based	Uniform	High PE, low precision on context
Adv. Beginner		Context-sensitive	Slightly shaped	Beginning to weight context
Competent		Goal-organized	Moderately shaped	Selective precision on goal-relevant signals
Proficient		Holistic pattern recognition	Strongly shaped	High precision on affordances
Expert		Intuitive, low PE	Habitual	Minimal deliberation, maximal automaticity

Now answer: At what stage is **free energy minimization most computationally intensive**? At what stage is it most efficient? What does this say about the relationship between expertise and cognitive effort?

Write your response here

Part 4: Epistemic Actions in Daily Life

Exercise: Identify three "epistemic actions" from your daily life — actions taken primarily to reduce uncertainty rather than achieve a goal. For each:

1. Describe the action
2. Identify the hidden state you were uncertain about
3. Compute the epistemic value: what uncertainty was reduced?

4. Was there also pragmatic value? If so, which dominated?

Example: Checking the weather app before dressing → Hidden state: weather. Epistemic value: reduces outfit decision uncertainty. Pragmatic value: avoids being cold/wet.

Write your response here

Part 5: Philosophical Debate — Active Inference and the Spectator Theory

In 250 words, engage with this debate: The **spectator theory of knowledge** says the mind passively observes the world. Pragmatists (Dewey especially) reject this — knowledge requires action. Active Inference formalizes this through the perception-action loop.

But does Active Inference's mathematical formalism reintroduce the spectator theory? After all, the generative model is a mathematical representation "inside" the agent, and perception is described as inference (a seemingly passive operation). How do you resolve this tension?

Consider:

- Does the fact that perception requires precision optimization (an active process) adequately address the concern?
- Is the action-perception loop genuinely different from a spectator who also acts, or is this a distinction without a difference?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Historical mapping	Pragmatism ↔ Active Inference
2	Applied analysis	Affordance formalization
3	Self-analysis	Expertise and precision dynamics
4	Daily observation	Epistemic actions and EFE
5	Philosophical debate	Spectator theory critique